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Table: 10.1 Ionic Channels in the Axolemma

Leakage Channels are ionic channels that remain open all the time	K^+ leakage channels are more in number than the Na^+ leakage channels. Sarcolemma has greater permeability to K^+ ions than Na^+ ions. These ions keep moving continuosly maintain the potential difference across the axolemma.
Ligand-gated Channels are chemically gated channels which open or close in response to a chemical stimuli.	They are located between the presynaptic membrane of the first axon and post synaptic membrane of the cell body of second neuron [i.e. dendrites and cell bodies]. The neurotransmitter acetylcholine opens ligand channels that allow Na^+ and Ca^{++} ions diffuse inward and K^+ ions diffuse outward.
Voltage-gated Channels are mechanically gated channels which open in response to a physical stimulus in the form of vibration such as touch and pressure.	These channels open in response to a change in membrane potential. There are two types of voltage-gated channels. **i. Sodium voltage-gated channel** **ii. Potassium voltage-gated channel**

neuronal cells. The ECF and ICF (cytosol) contains negatively charged particles (anions) and positively charged particles (cations). These charged particles are involved in the conduction of impulses.

The neurons maintain an uneven distribution of various inorganic ions across their axolemma for transmission of impulses. This unequal distribution of ions establishes the membrane potential across the axolemma. The axolemma contains a variety of membrane proteins that act as ionic channels and regulates the movement of ions across the axolemma. (Shown in Table 10.1).

10.3.2 Transmission of impulses

The transmission of impulse involves two main phases; **Resting membrane potential** and **Action membrane potential.**

Resting membrane Potential: The electrical potential difference across the plasma membrane of a resting neuron is called the **resting potential** during which the interior of the cell is negative due to greater efflux of K^+ outside the cell than Na^+ influx into the cell. When the axon is not conducting any impulses i.e. in resting condition, the axon membrane is more

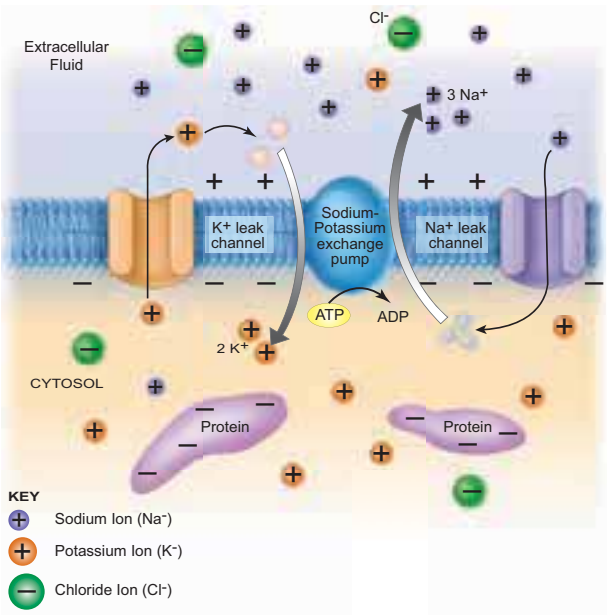


Figure 10.3 Ionic channels